ALWAYS ONE DIRECTION

Logarithmic functions are strictly monotonic: either always increasing or always decreasing throughout their domain. The base decides which.

MONOTONICITY

$$f(x) = \log\ (x)$$

split by **the base a**

↑

CASE 1

$a > 1$

Strictly increasing

larger inputs produce larger outputs

$$x\ < x\ \Rightarrow \log\ (x\ ) < \log\ (x\ )$$

E.G. $\log\ (x), \log\ \ (x), \ln(x)$

1 2

2 10

↓

CASE 2

$0 < a < 1$

Strictly decreasing

larger inputs produce smaller outputs

$$x\ < x\ \Rightarrow \log\ (x\ ) > \log\ (x\ )$$

1 2 E.G. $\log\ \ (x)$ a 2

1/2

TAKEAWAY

This is why taking logarithms preserves inequality direction when $a > 1$ and reverses it when $0 < a < 1$; see **inequalities**.